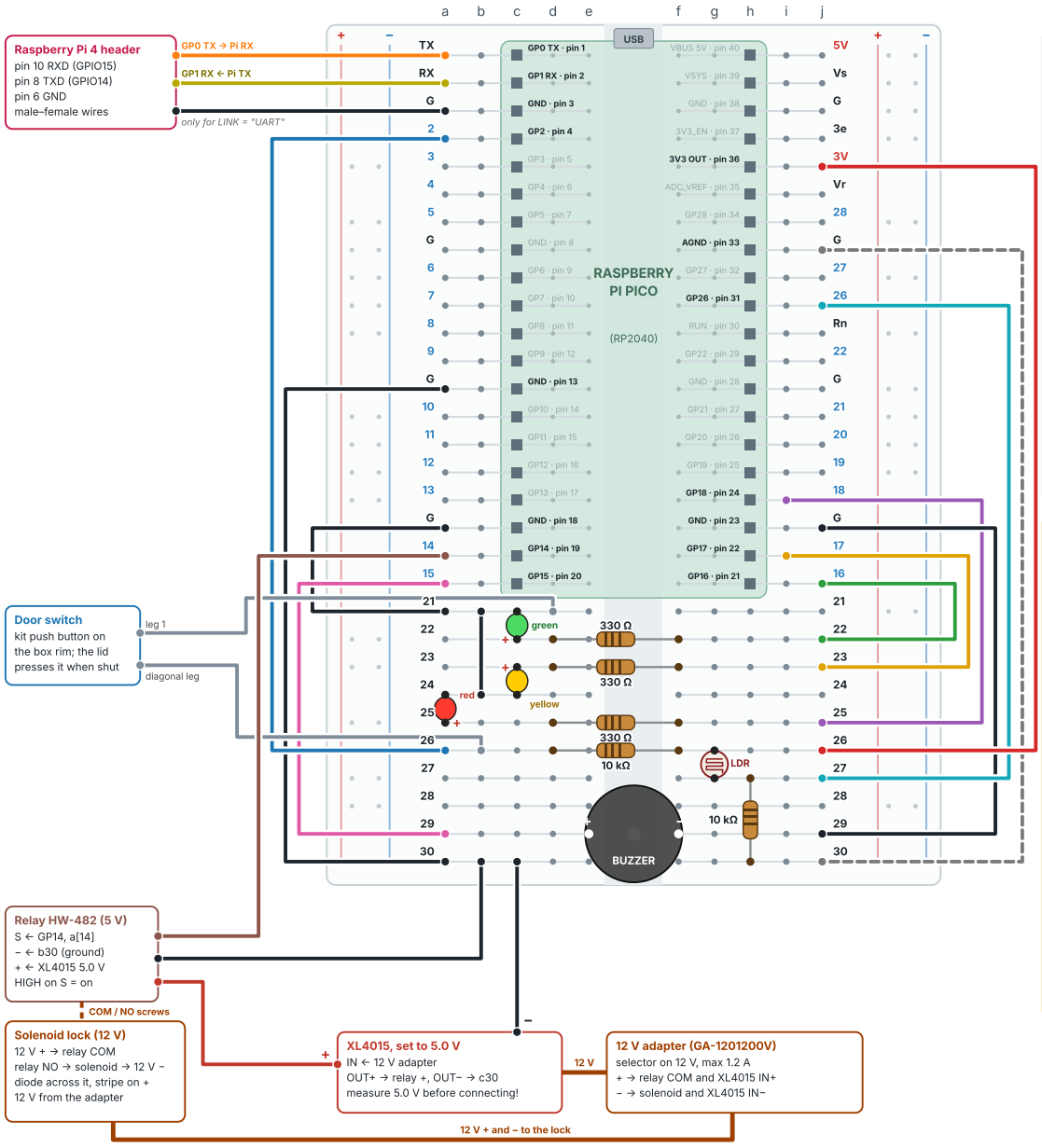
Team 17 — Pico breadboard wiring reference

MonkMakes Pico BB · Pico with USB at the top, pins in columns c and h · labels at the row ends are exactly what the board prints
Beside the Pico, a hole is named by column + printed label in brackets: a[2] = column a, row printed 2 (GP2). Below the Pico, plain numbers: a26 = column a, row 26.
Inside the Pico: pin name · Pico pin number. The + / − rails on both sides are not used in this layout.



Wire colours

- Ground (GND)
- Analogue ground (AGND)
- 3.3 V from the Pico
- 5 V from the XL4015 (off board)
- 12 V adapter (off board)
- GP16 → green LED
- GP17 → amber LED
- GP18 → red LED
- GP2 ← door switch
- Wires to the push button
- GP26 ← light sensor
- GP15 → buzzer
- GP14 → relay S
- GP0 TX → Pi pin 10
- GP1 RX ← Pi pin 8

Parts

3 × LED + 330 Ω (+ = long leg)
10 kΩ door pull-up, d26–f26
LDR g26–g27 + 10 kΩ h27–h30
Passive buzzer e29(+)/f29(−)

Before you start
Unplug the Pico USB and the 12 V adapter while you move wires. Set the XL4015 to 5.0 V with a multimeter first.

Never connect
5 V to any Pico row. Rows printed 5V, Vs, 3e, Vr and Rn: leave empty.

Grounds
Ground rows: 21, 24 and 30 on the left, 29 on the right. Row 30 right is AGND, only for the light sensor. It is fed from j[G] between 28 and 27 = AGND.

Buzzer
Its round body covers the middle holes of rows 27–30. Keep them free.

ESP8266 HUZZAH
Own small breadboard, USB power only. No wire to the Pico or the Pi — on purpose.

Lock
Relay switches itself off after 3 s, so the solenoid can't overheat.